Round 2 research notes

CW and ASM International | Prepared 3 November 2026 | Prices: 30 October 2026

Scope follows the README in 6_Semiconductors_and_Aerospace_Round2.zip. Both the required CW row and optional ASM row are included. The workbook contains reported inputs and transparent calculations. It does not select investments, supply conclusions or provide price targets.

What is filled

Valuation inputs use FY2025 for both companies. Red flags cover FY2025 and FY2024, with FY2023 closing assets as the starting balance for FY2024. Both red-flag rows are complete for each company. Amounts are millions of the stated currency; shares are millions; rates are decimals. The original template structure is retained.

Cash, debt and outstanding shares all use 31 December 2025. The latest reported incremental dilution is added to that outstanding share count: Q2 2026 for CW and Q3 2026 for ASM. This follows the requested convention but is not an exact same-day fully diluted count. The component values and dates are visible in the workbook.

Details that affect interpretation

CW: SG&A combines the reported selling and general-and-administrative lines. Depreciation uses the separately reported PPE depreciation, rounded by the company. FY2024 long-term debt includes the finance lease amount classified in Other liabilities, avoiding omission from the borrowings-only balance.

ASM: reported D&A excludes impairment charges. Corporate cash excludes the restricted amount. The software column contains separately disclosed software additions on an accrual basis; the cash-flow statement combines software with other intangible purchases. Capitalized development is disclosed separately in the notes and is not software.

Guidance: CW uses the midpoint of its numeric FY2026 revenue range. Its Q3 results are scheduled for 4 November, so Q2 is the latest available release on this research date. ASM supplies numeric FY2026 growth at constant currencies; that basis is explicitly retained. Its FY2027 lower-bound outlook is documented separately.

Consensus: all three annual growth rates come from each company's single MarketScreener annual table, retrieved 3 November. They are not represented as an archived 30 October snapshot. Provider betas are also current checks, not archived date-specific estimates.

Yield: the U.S. 10-year is 4.94% on 30 October, verified against Treasury and consistent with the team convention. The official German Bund factsheet shows the 30 October reference yield of 3.34%; market quote services may show a different intraday or closing benchmark.

Price-history delivery

Four original Yahoo CSVs are included unchanged: CW, ASM.AS, SPY and EURUSD=X. CW and ASM were downloaded on 3 November; SPY and EURUSD were reused byte-for-byte from the 2 November download. The manifest records source URLs, dates, coverage and file hashes. Any incomplete November row remains in the original but is excluded from calculations.

Risk calculations and open fields

CW: worst available complete year 2008, USD adjusted-close return -32.9135%. Checked 2026-11-03. Worst complete year within Yahoo Max available history: 1986-2025. Adjusted-close return, including Yahoo dividend/split adjustments; not drawdown. Yahoo CSV begins January 1985 although the Max UI start parameter is earlier. Earlier listing history is outside this dataset.

ASM.AS: worst available complete year 2008, USD adjusted-close return -64.7884%. Checked 2026-11-03. Worst complete year within Yahoo Max available history: 2000-2025. Adjusted-close return, including Yahoo dividend/split adjustments; not drawdown. Yahoo ASM begins January 1999; first complete December-to-December year is 2000. FX: Yahoo EURUSD=X from December 2003; 1999-2002 December endpoints use the last published daily Federal Reserve DEXUSEU observation. FX timing differs (Federal Reserve noon versus Yahoo close). Earlier Amsterdam listing history is outside this dataset.

Annual return = (current December adjusted close × current December USD exchange rate) / (previous December adjusted close × previous December USD exchange rate), minus 1. Risk-tab formulas are supported by December_endpoints.csv and Annual_USD_returns.csv. This measures a calendar-year return, not peak-to-trough drawdown. Earlier exchange history outside Yahoo Max has not been claimed as covered.

Every field still MISSING

1. CW - capex_software

CW does not separately quantify FY2025 software cash capex. Expanded Yahoo Gold cash flows also show no software-only line and a blank intangibles-purchases figure. No standalone software fact was found in the SEC tagged data. A blank is not zero. Importance: Moderate; exact FCF scope.

2. CW - interest_income

CW bundles interest income with pension and other items in Other income, net. Yahoo Gold has no standalone figure. Macroaxis labels USD 43.15m as income, but this matches the rounded reported interest expense; that candidate is rejected. Importance: Moderate; does not block EBIT-based DCF inputs.

3. ASM.AS - total_discounted_lease_liability

ASM reports EUR 13.9m current leases and EUR 19.6m noncurrent contractual obligations, but the latter is not verified as discounted carrying value. StockAnalysis, FT and Yahoo show conflicting or incomplete totals. The reviewed ESEF package has no separate carrying-value fact. Importance: Supplementary; does not block debt under this README.

4. ASM.AS - software_only_cash_outflow

ASM reports EUR 44.5m software additions and EUR 44.7m cash purchases of all intangibles. Yahoo Gold combines capex and does not provide a software-only cash split. The additions figure is verified on its stated basis only. Importance: Moderate; FCF interpretation.

Two main numerical inputs remain blank: CW software capex and interest income. The two ASM gaps are supplementary disclosure checks. Company IR or an audited reconciliation is the next route for the four amounts. Remaining_MISSING.csv records each impact and possible source in full. All requested red-flag and risk cells remain filled.

Follow-up verification

Further search completed 3 November 2026. The original five gaps now consist of one resolved listing and four unverified financial amounts. Main numerical inputs have not been replaced with provider guesses.

WInS listing resolved

Within the active 2026-2027 WGHS Investment Competition, select Europe, then Amsterdam. The symbol search returns ASM INTERNATIONAL; selecting it loads ASM International NV, symbol ASM, with a EUR quote. No order was reviewed or submitted. ASM.AS is the Yahoo symbol and failed in the generic quote service; that failure does not imply an unavailable Amsterdam listing.

WInS Trade Equities - checked 3 November 2026

Candidate values and reconciliation

CW, Interest income: 43.15 USD m, FY2025. REJECTED candidate. Macroaxis labels income; matches reported interest expense 43.148m.

Macroaxis - checked 3 November 2026

ASM.AS, Lease total: 77.9 EUR m, 31 Dec 2025. NOT FULLY VERIFIED. StockAnalysis: 13.9 current + 64.0 long-term; 64.0 matches all Other liabilities.

StockAnalysis - checked 3 November 2026

ASM.AS, Lease total: 34 EUR m (rounded), 31 Dec 2025. NOT FULLY VERIFIED. FT total debt; no reconciliation to discounted lease carrying values.

Financial Times - checked 3 November 2026

ASM.AS, Lease total: 13.9 EUR m, 31 Dec 2025. INCOMPLETE total. Yahoo total equals current leases; long-term FY2025 amount is blank.

Yahoo Finance Gold - checked 3 November 2026

ASM.AS, Software additions: 44.5 EUR m, FY2025. VERIFIED additions; cash NOT FULLY VERIFIED. Note 6 p179 additions. This is the existing software-column basis.

2025 annual report - checked 3 November 2026

ASM.AS, Intangible cash purchases: 44.7 EUR m, FY2025. VERIFIED aggregate; software split MISSING. Cash-flow p163; all intangibles, excluding separately reported development.

2025 annual report - checked 3 November 2026

Yahoo Gold aggregate ASM capex is EUR 468.3m: PPE 218.5m + capitalized development 205.1m + cash purchases of other intangibles 44.7m. This verifies the aggregate but does not isolate software cash payments. Source: Yahoo ASM cash-flow statement and annual report page 163.

Yahoo ASM cash flow - checked 3 November 2026

AI-use log and source access

Date: 3 November 2026. Tool: OpenAI Codex, with web research, local spreadsheet processing and read-only Yahoo Finance / WInS browser access. Prior native SPY and EURUSD downloads were made on 2 November and reused unchanged.

Work performed

Read the supplied README and template; reviewed company filings, guidance, consensus, yields and market histories; populated the template and calculated USD annual returns. Follow-up: expanded Yahoo Gold statements, checked SEC tags and conflicting provider figures, and verified ASM in WInS under Europe > Amsterdam > ASM. Added a dated gap review without substituting unverified numbers.

Verification performed

Checked financial-year mappings and share dates; reconciled cash and dilution arithmetic; checked that finance leases were not duplicated; recalculated workbook formulas; compared exported values with the reviewed data; inspected rendered workbook sections; verified original CSV hashes.

Human work still required

Open the cited filings and confirm the intended accounting mappings. Decide how to handle the four explicitly missing financial amounts and obtain IR clarification if needed. WInS listing presence is verified; no order or execution was tested. Log which AI research was used. Write all investment judgments and conclusions independently.

Disclosure

This package is AI-assisted research. It is not a student-authored submission. No emails, orders, purchases or investment transactions were made.

Primary source links

CW FY2025 10-K, filed 12 February 2026

CW Q2 2026 10-Q, filed 6 August 2026

ASM FY2025 annual report, published 12 March 2026

ASM Q3 results, 27 October 2026

German Finance Agency: 10-year Bund reference yield

Treasury daily yield curve, 2026

Federal Reserve / FRED euro exchange rate

All sources checked 3 November 2026. Source_ledger.csv provides the field-level URLs and period notes; the workbook contains the same links in its source columns and cell notes. The Yahoo manifest gives every download URL.